KAIZEN MATHS

# Lesson starter

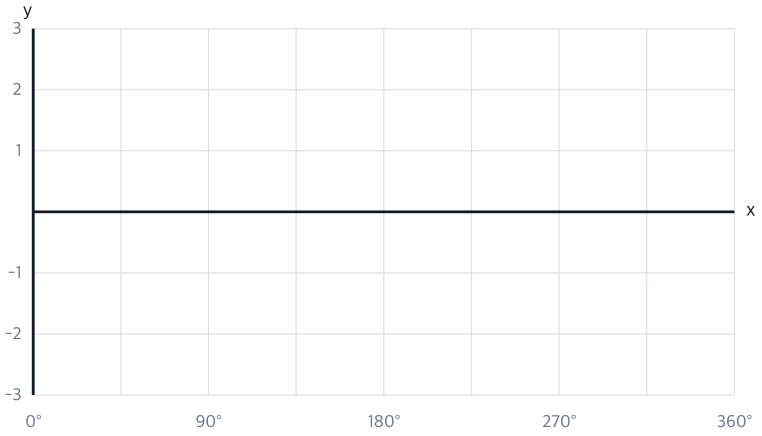
**Trigonometric Graphs and Transformations**

Tangent and Inverse Reasoning

Name _____

**SA1**

Sketch each tangent graph, showing asymptotes and intercepts.




Sketch y = tan(x + 45°) + 1 and show the vertical asymptotes.



Show your working

KAIZEN MATHS

Lesson starter

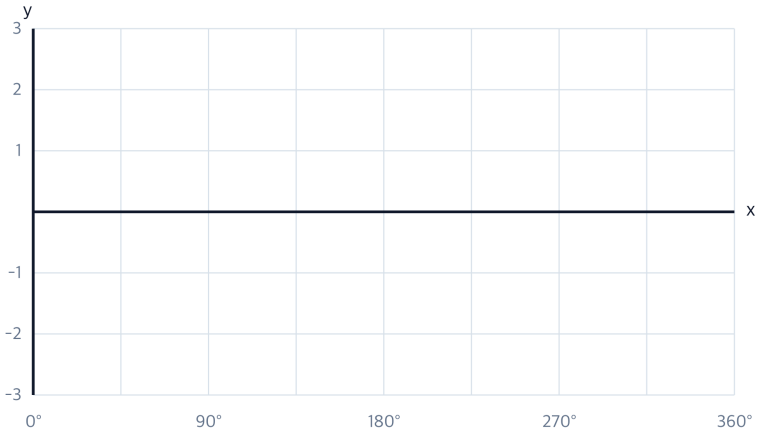
Trigonometric Graphs and Transformations

Tangent and Inverse Reasoning

Name _____

**SB1**

Find the asymptotes before sketching the tangent graph.




For y = tan(2(x - 30°)), find the vertical asymptotes in 0° ≤ x ≤ 360°, then sketch the graph.



Show your working

KAIZEN MATHS

# Lesson starter

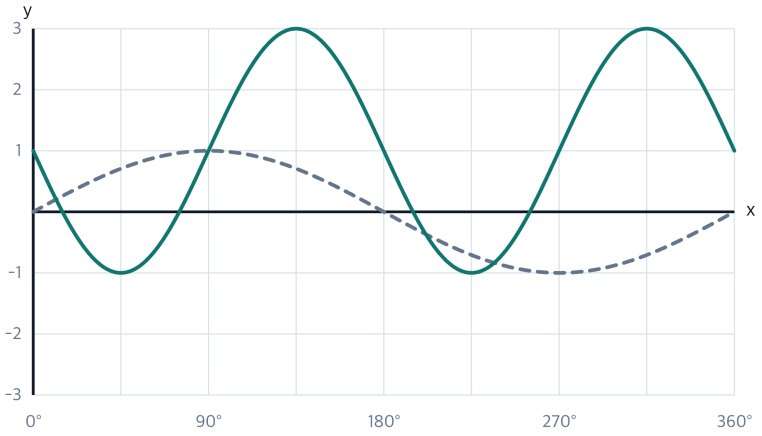
**Trigonometric Graphs and Transformations**

Tangent and Inverse Reasoning

Name _____

**SC1**

Compare the two graphs by their amplitude, period, midline, and phase shift.




dashed: parent graphsolid: transformed graph

Compare the dashed parent graph with the solid transformed graph. State the transformations.

ALG-020E-L4 · SC

Show your working

KAIZEN MATHS

Lesson starter

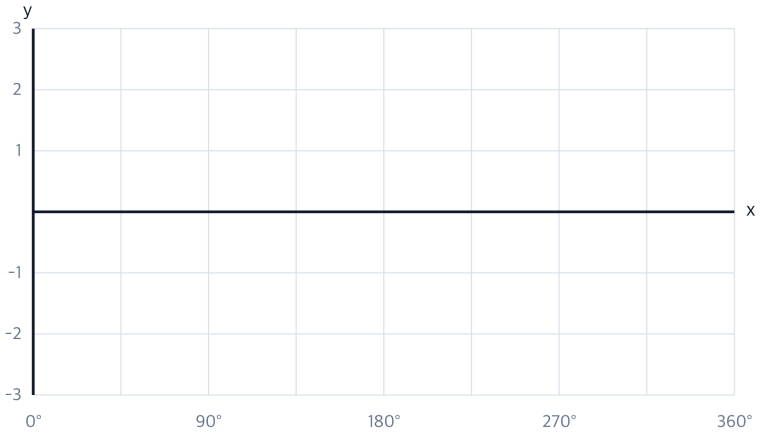
**Trigonometric Graphs and Transformations**

Tangent and Inverse Reasoning

Name _____

**SD1**

Sketch each tangent graph, showing asymptotes and intercepts.




Sketch y = tan(2(x)) - 1 and show the vertical asymptotes.

ALG-020E-L4 · SD

Show your working

KAIZEN MATHS

# Exit ticket

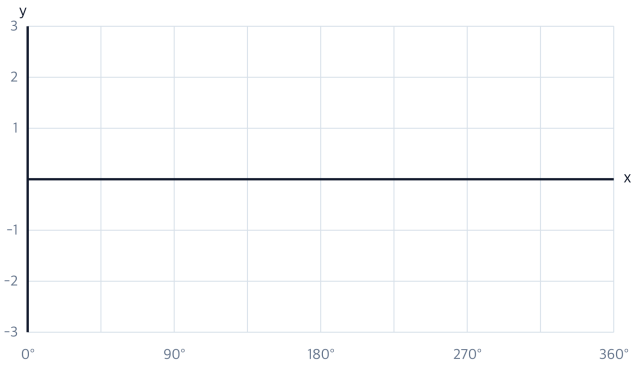
**Trigonometric Graphs and Transformations**

Tangent and Inverse Reasoning

Name

**EA1**

Find the asymptotes before sketching the tangent graph.




For y = tan(2(x + 30°)), find the vertical asymptotes in 0° ≤ x ≤ 360°, then sketch the graph.



Show your working

KAIZEN MATHS

# Exit ticket

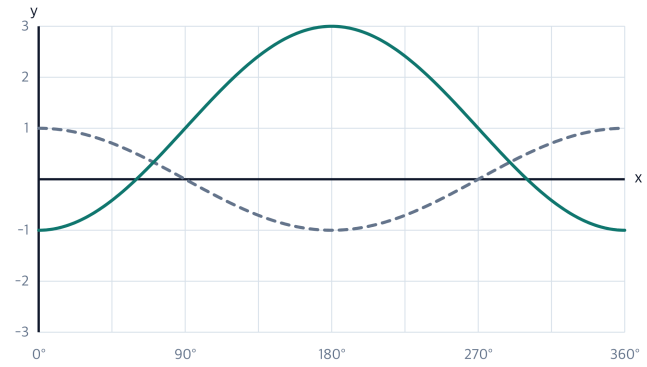
**Trigonometric Graphs and Transformations**

Tangent and Inverse Reasoning

Name

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Compare the two graphs by their amplitude, period, midline, and phase shift.




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Show your working

KAIZEN MATHS

# Exit ticket

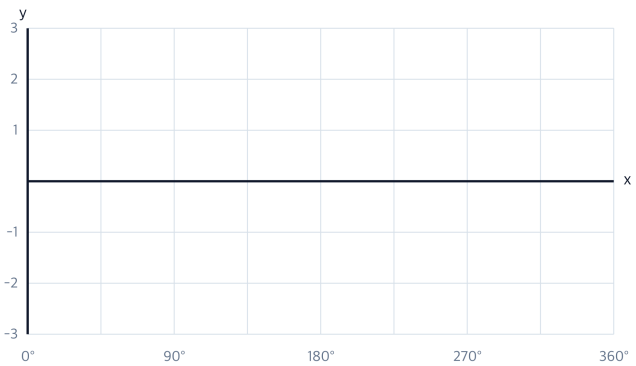
**Trigonometric Graphs and Transformations**

Tangent and Inverse Reasoning

Name

**EC1**

Sketch each tangent graph, showing asymptotes and intercepts.




Sketch y = tan(x + 45°) - 1 and show the vertical asymptotes.



Show your working

KAIZEN MATHS

# Exit ticket

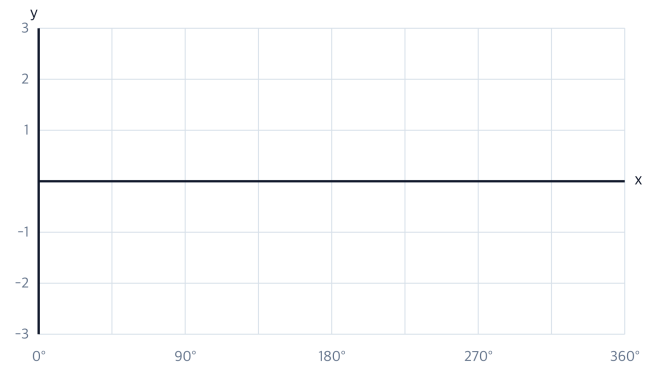
**Trigonometric Graphs and Transformations**

Tangent and Inverse Reasoning

Name

**EA1**

Find the asymptotes before sketching the tangent graph.




For y = tan(2(x + 30°)), find the vertical asymptotes in 0° ≤ x ≤ 360°, then sketch the graph.



Show your working

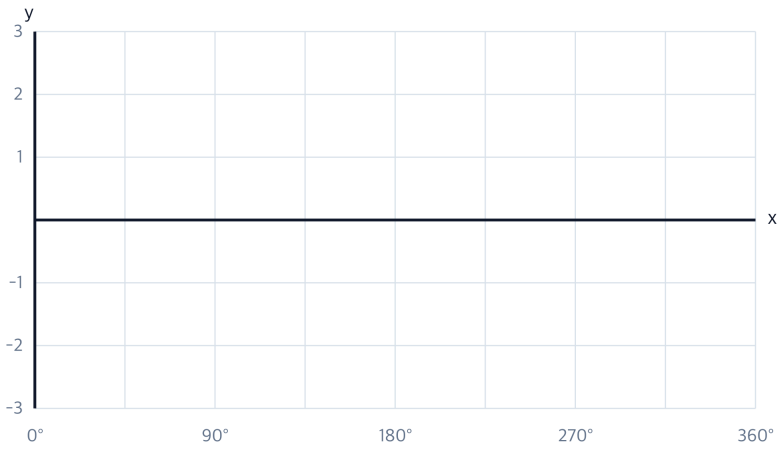
# Teacher answers & working

**Trigonometric Graphs and Transformations · Tangent and Inverse Reasoning**

ALG-020E-L4 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

## SA1

Sketch each tangent graph, showing asymptotes and intercepts.




Sketch y = tan(x + 45°) + 1 and show the vertical asymptotes.

**Answer:**

Period 180°; midline y = 1; asymptotes at 45°, 225°.

**Working:**

Start from the parent graph y = tan x.

The coefficient outside the trigonometric function is 1, so there is a vertical scale factor, but no amplitude for tangent.

The coefficient inside the bracket is 1, so the period is 180° ÷ 1 = 180°.

The graph is shifted 45° left.

The + 1 outside the function gives the midline y = 1.

Summarise the graph features.

| | |
|---|---|
| Amplitude | not defined for tangent |
| Period | 180° |
| Midline | $y = 1$ |
| Phase shift | 45° left |

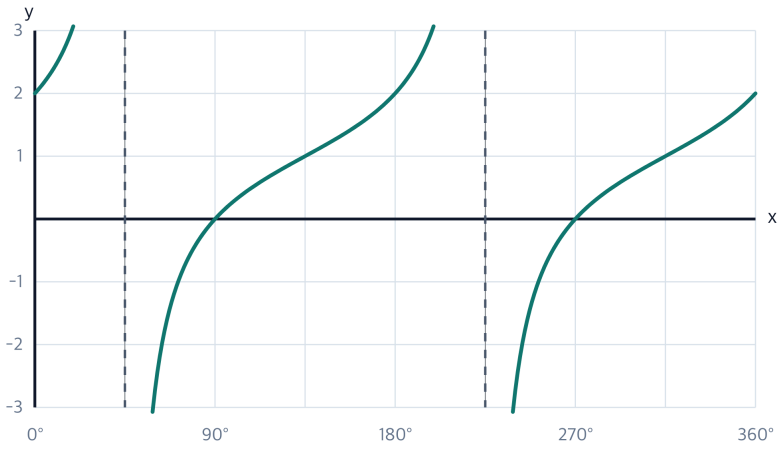
For tangent, mark the vertical asymptotes before drawing the branches: 45°, 225°.

# Teacher answers & working

**Trigonometric Graphs and Transformations · Tangent and Inverse Reasoning**

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## SA1 (continued)



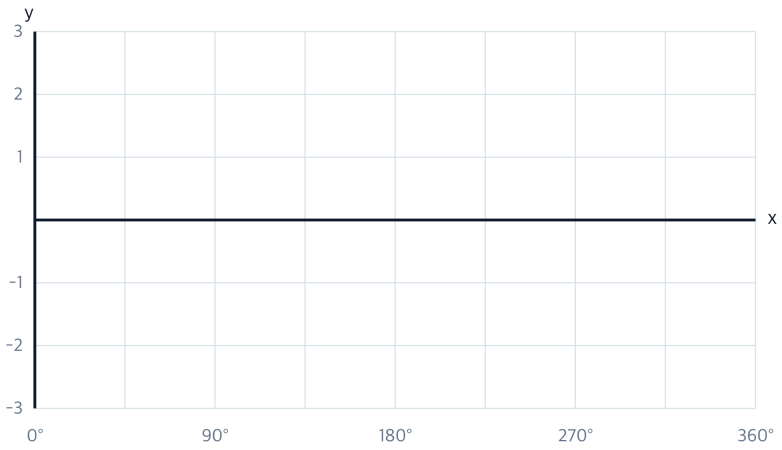
# Teacher answers & working

**Trigonometric Graphs and Transformations · Tangent and Inverse Reasoning**

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## SB1

Find the asymptotes before sketching the tangent graph.




For y = tan(2(x - 30°)), find the vertical asymptotes in 0° ≤ x ≤ 360°, then sketch the graph.

**Answer:**

Asymptotes: 75°, 165°, 255°, 345°.

**Working:**

Start from the parent graph y = tan x.

The coefficient outside the trigonometric function is 1, so there is a vertical scale factor, but no amplitude for tangent.

The coefficient inside the bracket is 2, so the period is 180° ÷ 2 = 90°.

The graph is shifted 30° right.

The + 0 outside the function gives the midline y = 0.

Summarise the graph features.

| | |
|---|---|
| Amplitude | not defined for tangent |
| Period | 90° |
| Midline | $y = 0$ |
| Phase shift | 30° right |

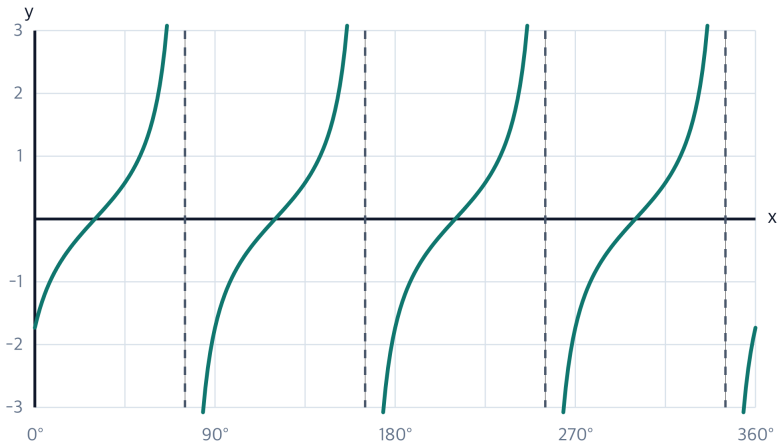
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# Teacher answers & working

**Trigonometric Graphs and Transformations · Tangent and Inverse Reasoning**

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## SB1 (continued)



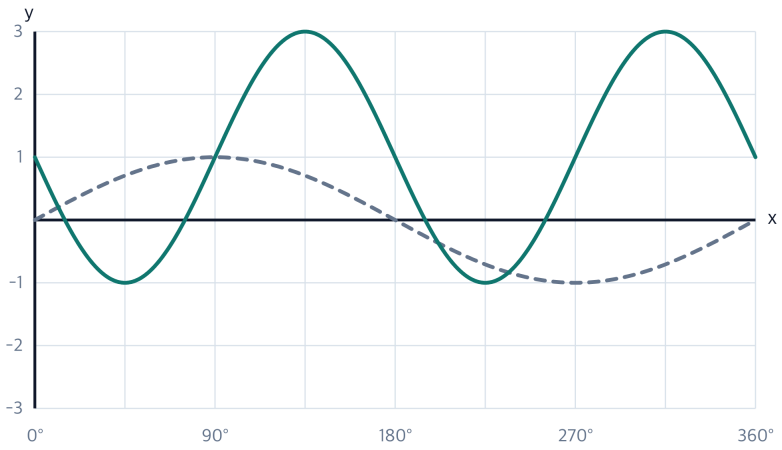
# Teacher answers & working

**Trigonometric Graphs and Transformations · Tangent and Inverse Reasoning**

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## SC1

Compare the two graphs by their amplitude, period, midline, and phase shift.




dashed: parent graphsolid: transformed graph

Compare the dashed parent graph with the solid transformed graph. State the transformations.

**Answer:**

Amplitude changes to 2, period becomes 180°, phase shift 90° right, midline y = 1.

**Working:**

Start from the parent graph y = sin x.

The coefficient outside the trigonometric function is 2, so the amplitude is |2| = 2.

The coefficient inside the bracket is 2, so the period is 360° ÷ 2 = 180°.

The graph is shifted 90° right.

The + 1 outside the function gives the midline y = 1.

Summarise the graph features.

| | |
|---|---|
| Amplitude | 2 |
| Period | 180° |
| Midline | y = 1 |
| Phase shift | 90° right |

Use the transformed key points to guide the sketch.

| x | y |
|---|---|
| 90° | 1 |
| 135° | 3 |
| 180° | 1 |
| 225° | -1 |
| 270° | 1 |

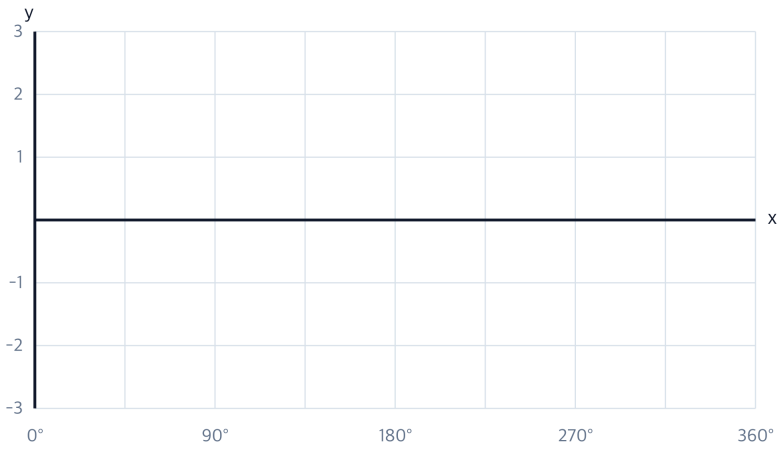
# Teacher answers & working

**Trigonometric Graphs and Transformations · Tangent and Inverse Reasoning**

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## SD1

Sketch each tangent graph, showing asymptotes and intercepts.




Sketch y = tan(2(x)) - 1 and show the vertical asymptotes.

**Answer:**

Period 90°; midline y = -1; asymptotes at 45°, 135°, 225°, 315°.

**Working:**

Start from the parent graph y = tan x.

The coefficient outside the trigonometric function is 1, so there is a vertical scale factor, but no amplitude for tangent.

The coefficient inside the bracket is 2, so the period is 180° ÷ 2 = 90°.

The graph is shifted 0° horizontally.

The - 1 outside the function gives the midline y = -1.

Summarise the graph features.

| | |
|---|---|
| Amplitude | not defined for tangent |
| Period | 90° |
| Midline | y = -1 |
| Phase shift | none |

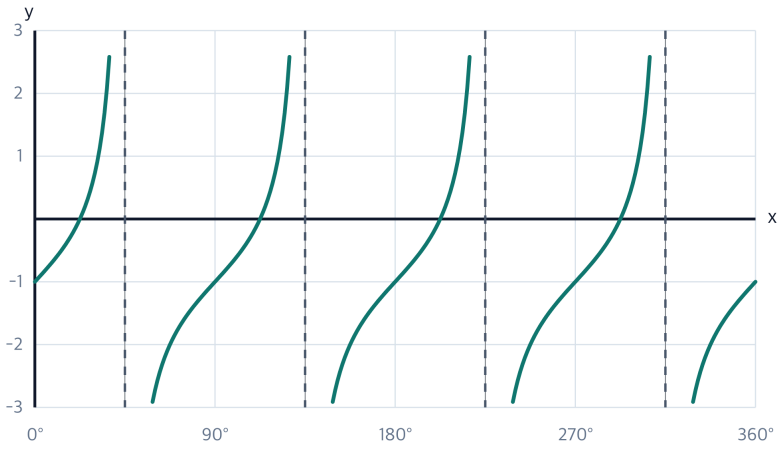
For tangent, mark the vertical asymptotes before drawing the branches: 45°, 135°, 225°, 315°.

# Teacher answers & working

**Trigonometric Graphs and Transformations · Tangent and Inverse Reasoning**

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## SD1 (continued)



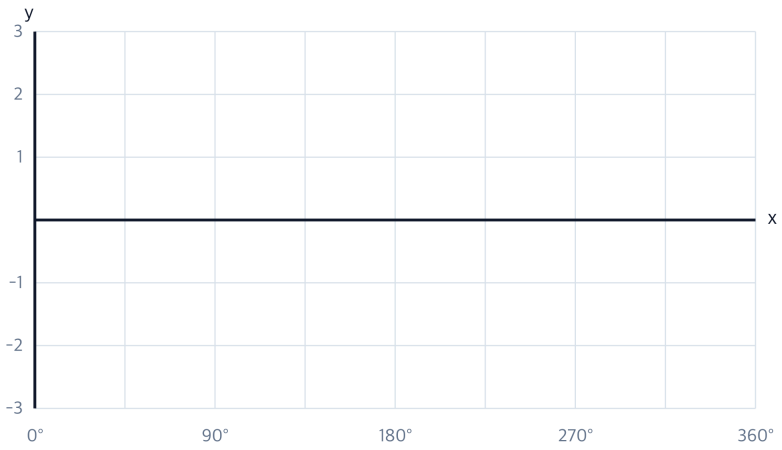
# Teacher answers & working

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## EA1

Find the asymptotes before sketching the tangent graph.




For y = tan(2(x + 30°)), find the vertical asymptotes in 0° ≤ x ≤ 360°, then sketch the graph.

**Answer:**

Asymptotes: 15°, 105°, 195°, 285°.

**Working:**

Start from the parent graph y = tan x.

The coefficient outside the trigonometric function is 1, so there is a vertical scale factor, but no amplitude for tangent.

The coefficient inside the bracket is 2, so the period is 180° ÷ 2 = 90°.

The graph is shifted 30° left.

The + 0 outside the function gives the midline y = 0.

Summarise the graph features.

| | |
|---|---|
| Amplitude | not defined for tangent |
| Period | 90° |
| Midline | $y = 0$ |
| Phase shift | 30° left |

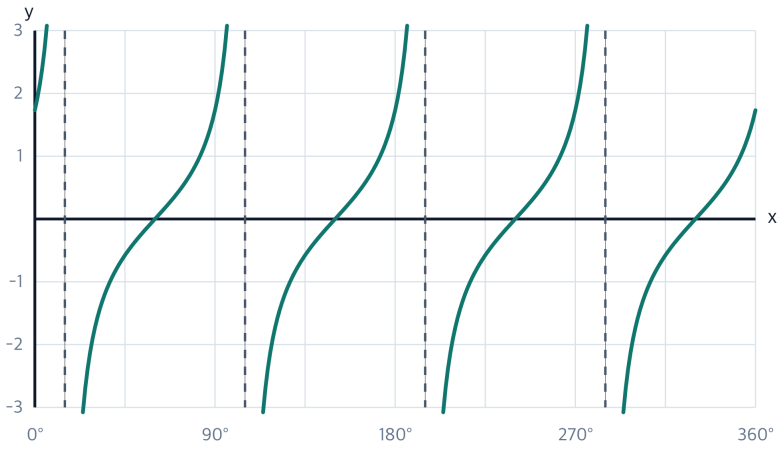
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# Teacher answers & working

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## EA1 (continued)



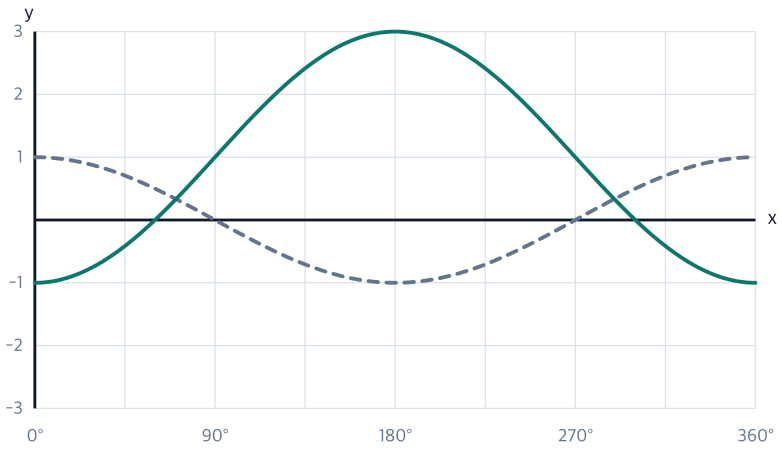
# Teacher answers & working

**Trigonometric Graphs and Transformations · Tangent and Inverse Reasoning**

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## EB1

Compare the two graphs by their amplitude, period, midline, and phase shift.




dashed: parent graphsolid: transformed graph

Compare the dashed parent graph with the solid transformed graph. State the transformations.

**Answer:**

Amplitude changes to 2, period becomes 360°, phase shift none, midline y = 1.

**Working:**

Start from the parent graph y = cos x.

The coefficient outside the trigonometric function is -2, so the amplitude is |-2| = 2.

The coefficient inside the bracket is 1, so the period is 360° ÷ 1 = 360°.

The graph is shifted 0° horizontally.

The + 1 outside the function gives the midline y = 1.

Summarise the graph features.

| | |
|---|---|
| Amplitude | 2 |
| Period | 360° |
| Midline | y = 1 |
| Phase shift | none |

Use the transformed key points to guide the sketch.

| x | y |
|---|---|
| 0° | -1 |
| 90° | 1 |
| 180° | 3 |
| 270° | 1 |
| 360° | -1 |

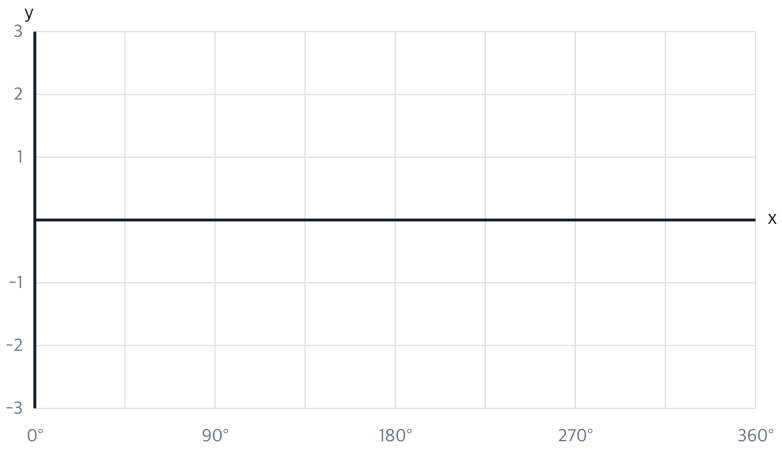
# Teacher answers & working

**Trigonometric Graphs and Transformations · Tangent and Inverse Reasoning**

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## EC1

Sketch each tangent graph, showing asymptotes and intercepts.




Sketch y = tan(x + 45°) - 1 and show the vertical asymptotes.

**Answer:**

Period 180°; midline y = -1; asymptotes at 45°, 225°.

**Working:**

Start from the parent graph y = tan x.

The coefficient outside the trigonometric function is 1, so there is a vertical scale factor, but no amplitude for tangent.

The coefficient inside the bracket is 1, so the period is 180° ÷ 1 = 180°.

The graph is shifted 45° left.

The - 1 outside the function gives the midline y = -1.

Summarise the graph features.

| | |
|---|---|
| Amplitude | not defined for tangent |
| Period | 180° |
| Midline | y = -1 |
| Phase shift | 45° left |

For tangent, mark the vertical asymptotes before drawing the branches: 45°, 225°.

# Teacher answers & working

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## EC1 (continued)

